

Light and Photosynthesis

Teacher guide | Biology 2e, Chapter 8 | 45-60 minutes

Learning targets and timing

How does light level affect the time it takes leaf disks to float?

Observe oxygen accumulation in infiltrated leaf disks. Compare three light conditions, repeat each treatment, and use flotation time as an indirect measure of net photosynthetic activity.

Suggested sequence: 8-10 min prediction and setup; 20-30 min observation; 12-20 min analysis. Virtual time can be accelerated. For photosynthesis and yeast, pool independent replicates from class groups; three sequential repeats need extra class time.

Route choices and sharing

The photosynthesis student page opens as a standard virtual investigation. Teacher customization is deferred to a future teacher workspace. Use this worksheet to assign classroom components separately; discuss data-source differences when comparing results. Existing hybrid links still work, but the student page no longer creates them.

Materials

- Fresh spinach leaves; hole punch; three clear cups; needle-free syringe
- 0.2% sodium bicarbonate solution (0.2 g per 100 mL water), trace of dish detergent
- LED light, foil or opaque cover, timer, thermometer; 10 disks per trial

Classroom procedure

1. Make equal-sized leaf disks, avoiding major veins. Prepare matched cups of solution and label dark, low light, and bright light.
2. In a needle-free syringe, immerse disks in solution, expel air, cover the tip, and gently draw back the plunger. Release and repeat until disks sink. Use the same preparation for every treatment.
3. Put 10 sunken disks in each cup. Use a cover for darkness and a shade screen for lower light; keep depth, leaf source, and temperature matched. The virtual percentages are relative settings, not calibrated classroom illuminance.
4. Count floating disks at 0, 2, 4, ... 20 minutes. Briefly observe the dark treatment with minimal illumination. Record counts even if they decrease or fail to reach five.
5. Repeat with new disks for trials 2 and 3. Independent cups, rather than individual disks in a shared cup, are the experimental replicates. Pool independent trials from class groups for a single-period lab; sequential repeats require additional time.

Safety: Use a needle-free syringe and a cool LED light. Keep electrical equipment dry. Do not taste solutions. Teacher supervision is needed for hole punches and glassware.

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Expected reasoning and troubleshooting

Virtual expectation: bright light generally reaches ET50 sooner than low light; darkness normally does not reach it during this model run. Exact times vary. Do not grade classroom results against the simulation.

Students should connect light-driven oxygen production and buoyancy, and recognize that respiration and leaf preparation influence net flotation.

If disks float at time zero, recheck infiltration. If none rise, check leaf condition, light, temperature, bicarbonate preparation, and excess detergent. Keep unexpected real observations; do not replace them with expected values.

Model boundaries

The virtual assay uses illustrative flotation thresholds with variation between trials. It is not calibrated to a particular leaf or lamp. Real disks can sink again; this simplified simulation models only their first rise.

ET50 is the interpolated time when 5 of 10 disks float. Lower ET50 suggests faster net oxygen accumulation under matched conditions. If five disks never float, report "not reached within 20 minutes"; do not treat that as a zero rate.

Suggested assessment - 12 points

Dimension	Evidence to look for	Points
Design	Reasoned prediction; variables, controls or sampling plan	0-3
Data	Complete observations, correct units, source labels, repeats	0-3
Analysis	Appropriate calculations and comparisons supported by data	0-3
Explanation	Biological reasoning, uncertainty, and a feasible improvement	0-3

For each dimension: 0 = absent; 1 = partial or major errors; 2 = mostly sound with a minor omission; 3 = complete and well supported. Suggested teacher rubric only; the app does not assign a grade.

Downloads, access, and classroom pilot

Student worksheet: three printable pages. Teacher guide: procedures, interpretation, and rubric. Reports and CSV data download from Analyze & explain. Teacher guides are public resources, not protected answer keys. Before classroom use, pilot the equipment, timing, and reading level. Editable Word and fillable PDF versions are planned.

Reading and method connections

OpenStax Biology 2e, 8.2: [Open source](#)

College Board, Biology Investigative Labs, Investigation 5: [Open source](#)

Text and diagrams in these labs are original teaching materials. Simulated observations are not measurements from the linked sources.